

Chapter1. Introduction

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Outline

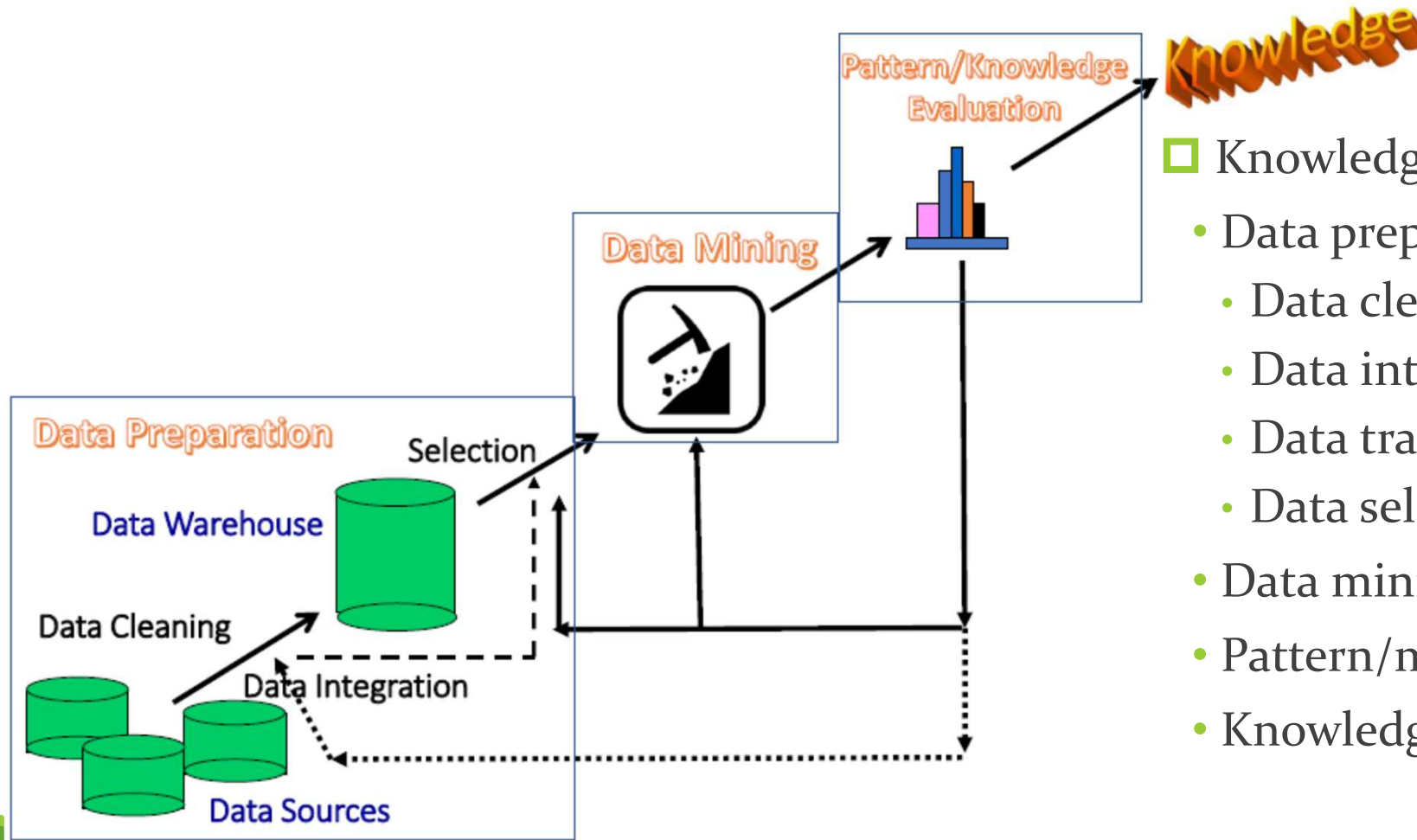
1. What is data mining?
 2. Data mining: an essential step in knowledge discovery
 3. What kinds of data can be mined?
 4. Mining various kinds of knowledge
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What is data mining?

- ❑ We live in a world where **vast amounts of data** are generated constantly and rapidly
- ❑ **Data mining** is the process of discovering interesting patterns, models and other kinds of knowledge in large data sets
 - “Data mining”: a misnomer? It should be “knowledge mining from data”
 - Other terms: *Knowledge mining from data, KDD (Knowledge Discovery from Data), pattern discovery, knowledge extraction, data analytics, information harvesting*
- ❑ Data mining is a young, dynamic, and promising field
- ❑ **Example: Data mining turns a large collection of data into knowledge**
 - Google’s *Flu Trends* found a close relationship between the number of people who search for flu-related info. and the number of people who have flu symptoms
 - It can estimate **flu activity** up to **two weeks faster than** traditional systems

Data Mining: An Essential Step in Knowledge Discovery



□ Knowledge Discovery Process

- Data preparation
- Data cleaning
- Data integration
- Data transformation
- Data selection
- Data mining
- Pattern/model evaluation
- Knowledge presentation

What kinds of data can be mined? (I)

□ Structured vs. unstructured data

- **Structured**: uniform, record- or table-like structures, defined by data dictionaries, with a fixed set of attributes, each with a fixed set of value ranges and semantic meaning
 - Ex. Data stored in *relational databases, data cubes, data matrices, and many data warehouses*
- **Semi-structured**: allow a data object to contain a set value, a small set of heterogeneous typed values, or nested structures, or to allow the structure of objects or sub-objects to be defined flexibly and dynamically (Ex. **JSON format**)
 - Data having *certain structures* with clearly defined semantic meaning, such as *transactional data set, sequence data set* (e.g., time-series data, gene or protein data, or Weblog data)
 - *Graph or network data*: A more sophisticated type of semi-structured data set
- **Unstructured data**: text data and multimedia (e.g., audio, image, video) data

□ The real-world data can often be a mixture of structured, semi-structured data and unstructured data

What kinds of data can be mined? (II)

□ Data associated with different applications

- Different applications: different data sets and require different data analysis methods
 - Sequence data: *Biological sequences vs. shopping transaction sequences*
 - Time-series: ordered set of numerical values with equal time interval
 - Spatial, temporal and spatiotemporal data
 - Graph and network data: Social networks, computer communication networks, biological networks, and information networks may carry rather different semantics
- On the same data set, finding different kinds of patterns: require different mining methods
 - Ex. software programs: finding plagiarized modules vs. finding copy-and-paste bugs

□ Stored vs. streaming data

- Stored data: Finite, stored in various kinds of large data repositories
- Streaming data (e.g., video surveillance or remote sensing): Dynamic, constantly coming, infinite, real-time response—posing challenges on effective data mining

Mining various kinds of knowledge

- ❑ Multidimensional data summarization
- ❑ Mining frequent patterns, associations, and correlations
- ❑ Classification and regression for predictive analysis
- ❑ Cluster analysis
- ❑ Deep learning
- ❑ Outlier analysis
- ❑ Are all mining results interesting?

Multidimensional data summarization

❑ Information integration and data warehouse construction

- Data cleaning, transformation, integration, and multidimensional data model

❑ Data cube technology

- Scalable methods for computing (i.e., materializing) multidimensional aggregates
- OLAP (online analytical processing)

❑ Multidimensional concept description: Characterization and discrimination

- Generalize, summarize, and contrast data characteristics, e.g., **dry vs. wet region**

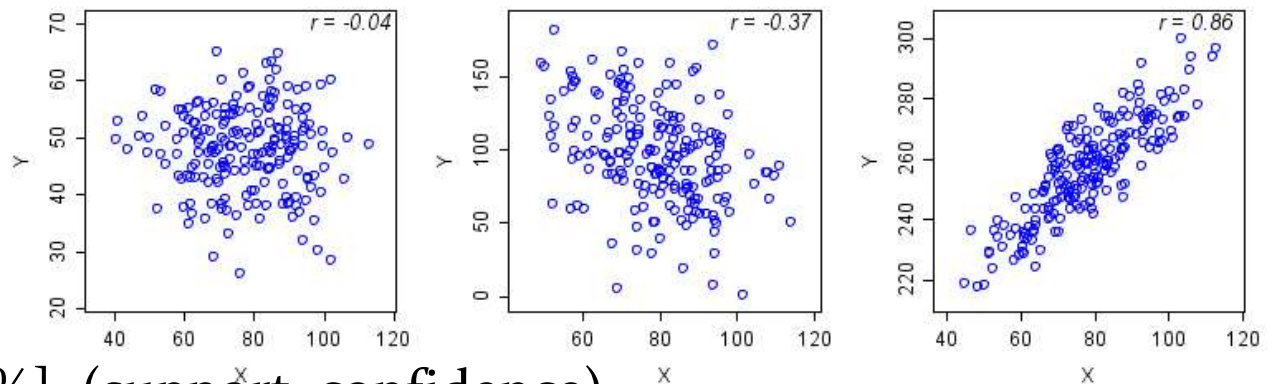


Pattern discovery: mining frequent patterns, associations, and correlations

□ Frequent patterns (or frequent itemsets)

- What items are frequently purchased together in your Walmart?

□ Association and Correlation Analysis



□ A typical association rule

- Diaper $\rightarrow$ Beer [0.5%, 75%] (support, confidence)
- Are strongly associated items also strongly correlated?

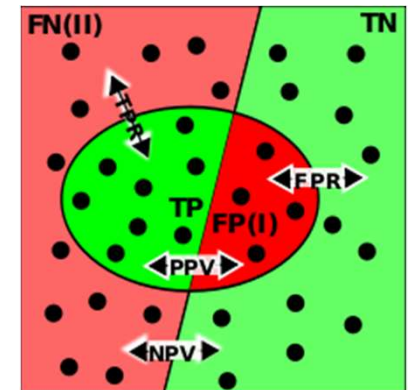
□ How to mine such patterns and rules efficiently in large datasets?

□ How to use such patterns for classification, clustering, and other applications?

Classification and regression for predictive analysis

□ Classification and label prediction

- Construct models (functions) based on some training examples
- Describe and distinguish classes or concepts for future prediction
 - Ex. 1. Classify countries based on (climate)
 - Ex. 2. Classify cars based on (gas mileage)
- Predict some unknown class labels



□ Typical methods

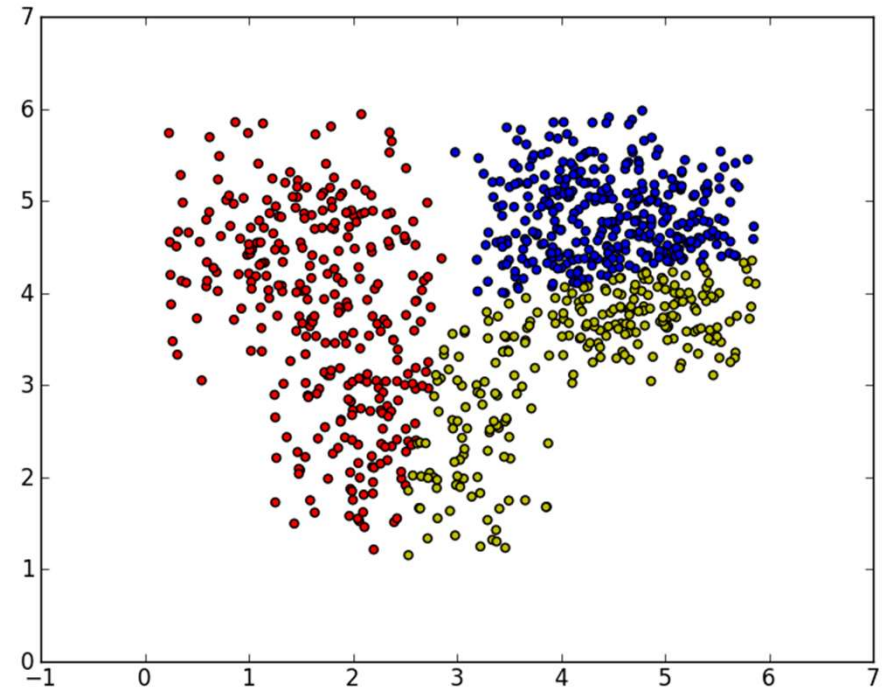
- Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, logistic regression, ...

□ Typical applications:

- Credit card fraud detection, direct marketing, classifying stars, diseases, web-pages, ...

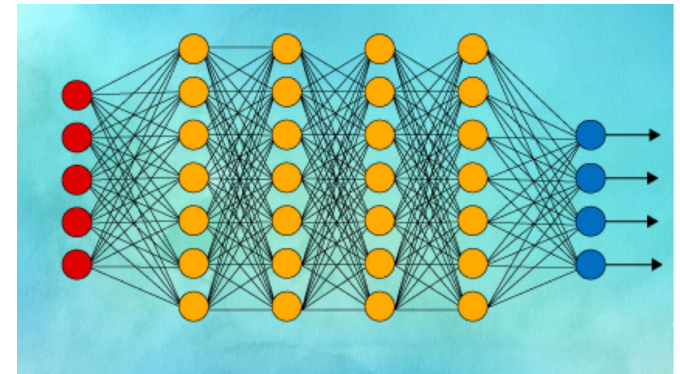
Cluster analysis

- Unsupervised learning (i.e., Class label is unknown)
- Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- Principle: Maximizing intra-class similarity & minimizing interclass similarity
- Many methods and applications



Deep learning

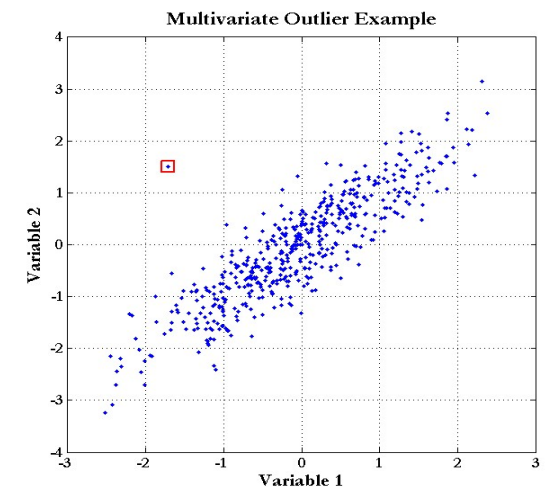
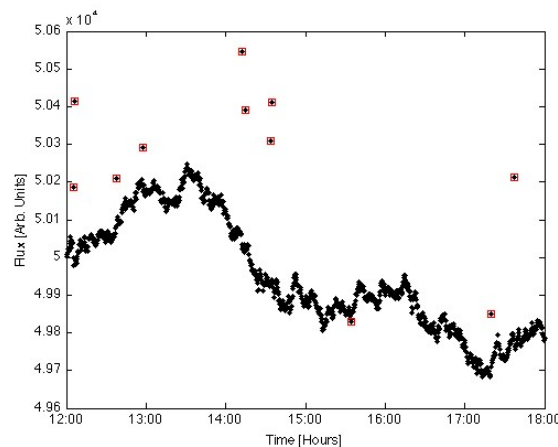
- ❑ Deep learning: A fast expanding dynamic frontier in machine learning
- ❑ Deep learning has developed various *neural network architectures*
 - Feed-forward neural networks
 - Convolutional neural networks
 - Recurrent neural networks
 - Graph neural networks
 - Transformer
- ❑ Deep learning has broad applications in computer vision, natural language processing, machine translation, social network analysis, and so on
- ❑ Deep learning has been reshaping a variety of data mining tasks
 - Ex. classification, clustering, outlier detection, and reinforcement learning



Outlier analysis

□ Outlier analysis

- Outlier: A data object that does not comply with the general behavior of the data
- Noise or exception?—One person's garbage could be another person's treasure
- Methods: by product of clustering or regression analysis, ...
- Useful in fraud detection, rare events analysis



Other data mining functions:

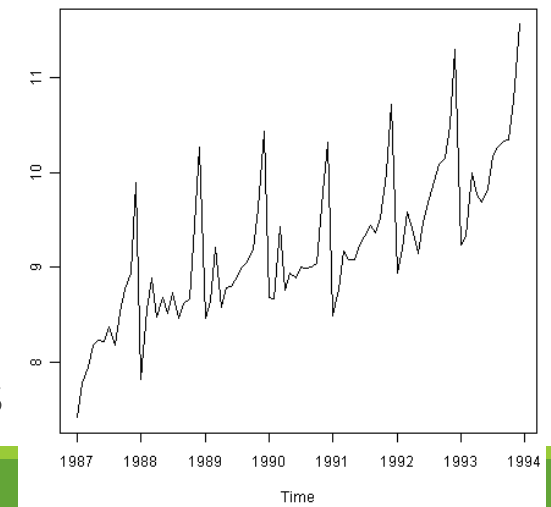
Time and ordering: sequential pattern, trend and evolution analysis

□ Sequence, trend and evolution analysis

- Trend, time-series, and deviation analysis
 - e.g., regression and value prediction
- Sequential pattern mining
 - e.g., buy digital camera, then buy large memory cards
- Periodicity analysis
- Motifs and biological sequence analysis
 - Approximate and consecutive motifs
- Similarity-based analysis

□ Mining data streams

- Ordered, time-varying, potentially infinite, data streams



Other data mining functions: structure and network Analysis

□ Graph mining

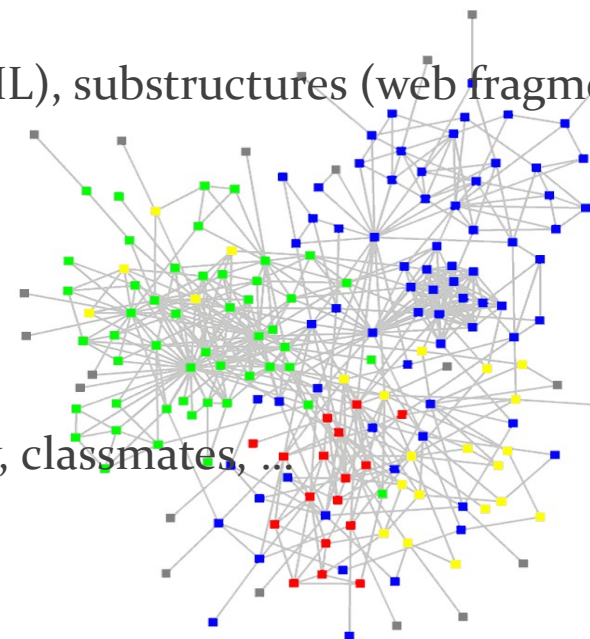
- Finding frequent subgraphs (e.g., chemical compounds), trees (XML), substructures (web fragments)

□ Information network analysis

- Social networks: actors (objects, nodes) and relationships (edges)
 - e.g., author networks in CS, terrorist networks
- Multiple heterogeneous networks
 - A person could be multiple information networks: friends, family, classmates, ...
- Links carry a lot of semantic information: Link mining

□ Web mining

- Web is a big information network: from PageRank to Google
- Analysis of Web information networks
 - Web community discovery, opinion mining, usage mining, ...

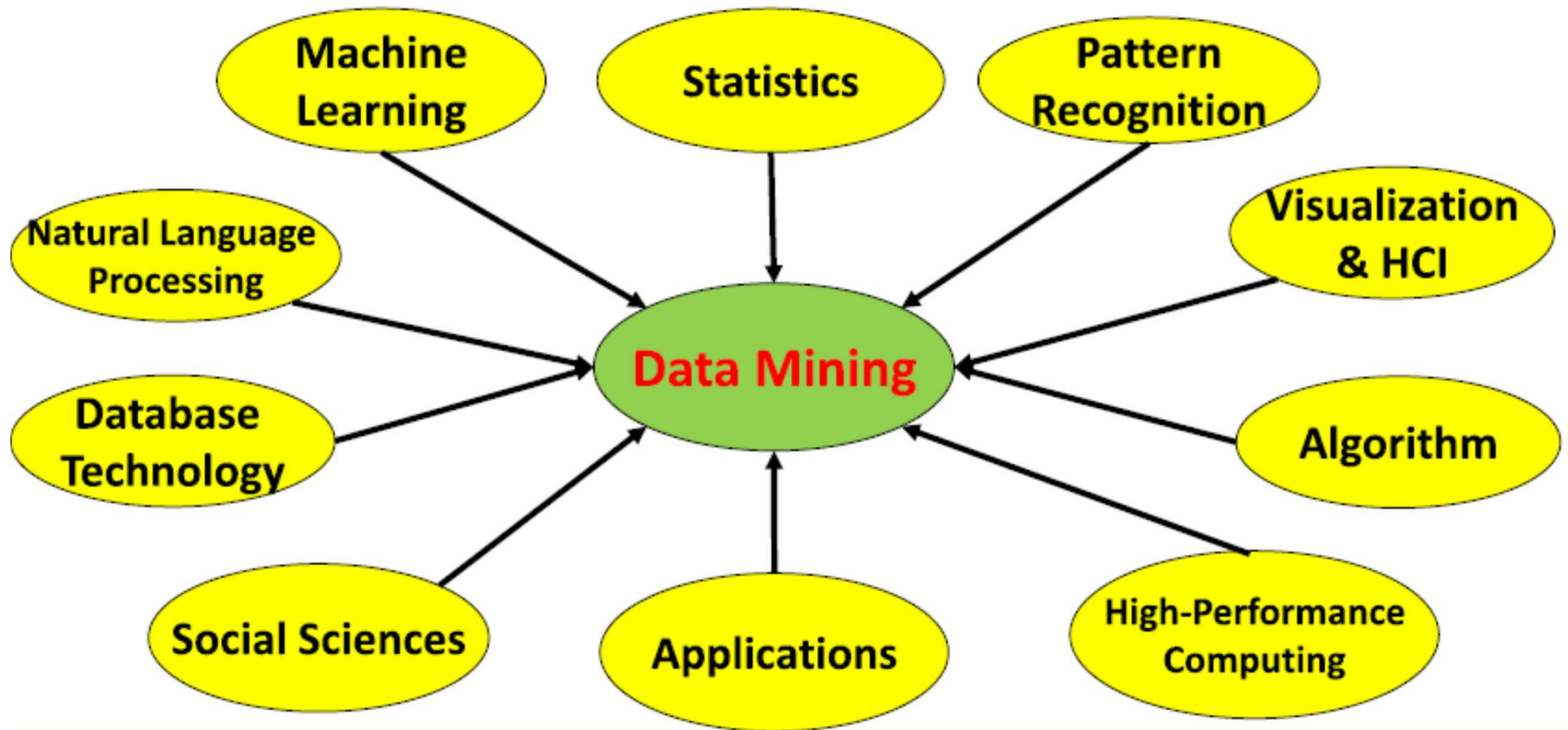


Evaluation of Knowledge

- ❑ Are all mined knowledge interesting?
 - One can mine tremendous amount of “patterns”
 - Some may fit only certain dimension space (time, location, ...)
 - Some may not be representative, may be transient, ...
- ❑ Evaluation of mined knowledge → directly mining only interesting knowledge?
 - Descriptive vs. predictive
 - Coverage
 - Typicality vs. novelty
 - Accuracy
 - Timeliness
 - ...



Data mining: confluence of multiple disciplines



Why confluence of multiple disciplines?

- Tremendous amount of data
 - Algorithms must be scalable to handle big data
- High-dimensionality of data
 - Micro-array may have tens of thousands of dimensions
- High complexity of data
 - Data streams and sensor data
 - Time-series data, temporal data, sequence data
 - Structure data, graphs, social and information networks
 - Spatial, spatiotemporal, multimedia, text and Web data
 - Software programs, scientific simulations
- New and sophisticated applications

What kinds of applications are targeted?

- ❑ Web page analysis: classification, clustering, ranking
- ❑ Collaborative analysis & recommender systems
- ❑ Basket data analysis to targeted marketing
- ❑ Biological and medical data analysis
- ❑ Data mining and software engineering
- ❑ Data mining and text analysis
- ❑ Data mining and social and information network analysis
- ❑ Built-in (invisible data mining) functions in Google, Microsoft, LinkedIn, Meta, ...
- ❑ Major dedicated data mining systems/tools
 - SAS, MS SQL-Server Analysis Manager, Oracle Data Mining Tools)



Data mining and society

- ❑ Data mining technology may benefit society
 - Ex.: Help scientific discovery, business management, economy recovery, and security protection (*e.g.*, the real-time discovery of intruders and cyberattacks)
- ❑ Need to guard against **the misuse of data mining**
 - Data mining also poses the risk of unintentionally disclosing some confidential business or government information and disclosing an individual's personal information
- ❑ Studies on **data security** in data mining and **privacy-preserving data** publishing and data mining are important, ongoing research theme
 - The philosophy is to observe data sensitivity and preserve data security and people's privacy while performing successful data mining
- ❑ These and other related issues will be discussed throughout the book

Summary

- ❑ Data mining: Discovering interesting patterns and knowledge from massive amounts of data
 - ❑ A KDD process includes data cleaning, data integration, data selection, transformation, data mining, pattern evaluation, and knowledge presentation
 - ❑ Different data mining method on a wide variety of data
 - ❑ Data mining functionalities: summarization, pattern discovery, classification, clustering, deep learning, outlier analysis, trend and outlier analysis, ...
 - ❑ Data mining is a confluence of multiple disciplines
 - ❑ Data mining has broad applications
 - ❑ Promote secure data mining to benefit society
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